Experiment 6 - Time Series Analysis of Website Traffic

Aim:

To analyze the time series data of website traffic to identify trends, seasonality, and anomalies.

Program:

```
import pandas as pd
import numpy as np
np.random.seed(42)
dates = pd.date_range(start='2025-01-01', periods=120, freq='D')
page_views = (
    5000
    + np.arange(120) * 15
    + 500 * np.sin(2 * np.pi * np.arange(120) / 7)
    + np.random.normal(0, 250, 120)
)
page_views[30] += 2500
page_views[75] -= 2000
page_views[100] += 3000
data = pd.DataFrame({
    'Date': dates,
    'Page_Views': page_views.astype(int)
})
data.to_csv('website_traffic.csv', index=False)
print(data.head(10))

data = pd.read_csv(
    'website_traffic.csv',
    parse_dates=['Date'],
    index_col='Date'
)
```

```
from statsmodels.tsa.seasonal import seasonal_decompose
import matplotlib.pyplot as plt


# Anomaly Detection
data['Page_Views_Z'] = (
    (data['Page_Views'] - data['Page_Views'].mean())
    / data['Page_Views'].std()
)

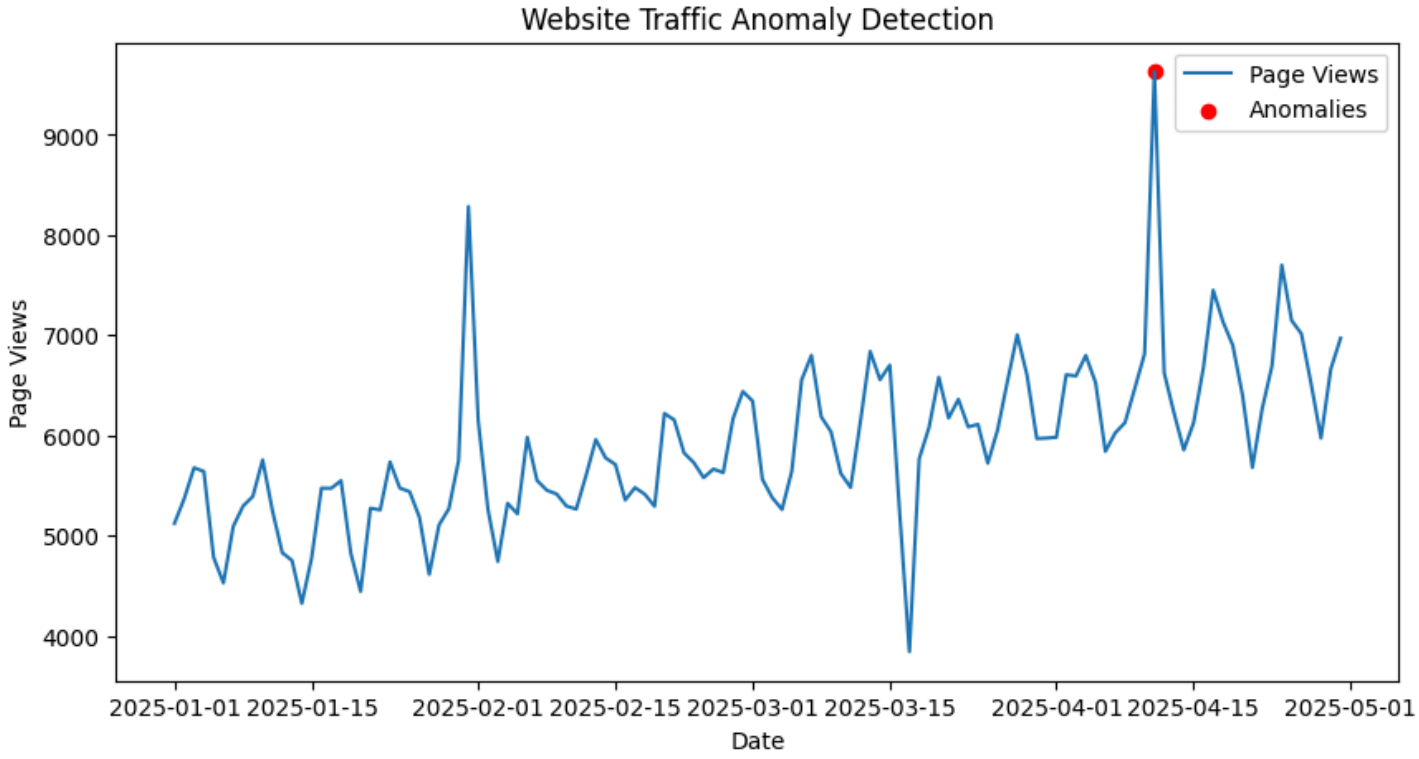
anomalies = data[data['Page_Views_Z'].abs() > 3]

plt.figure(figsize=(10, 5))
plt.plot(data.index, data['Page_Views'], label='Page Views')

plt.scatter(
    anomalies.index,
    anomalies['Page_Views'],
    color='red',
    label='Anomalies'
)

plt.title('Website Traffic Anomaly Detection')
plt.xlabel('Date')
plt.ylabel('Page Views')
plt.legend()
plt.show()
```

Output:



Result:

Thus, the time series data for website traffic anomaly detection has been analyzed.